

N-Shot Learning & Prompt Engineering

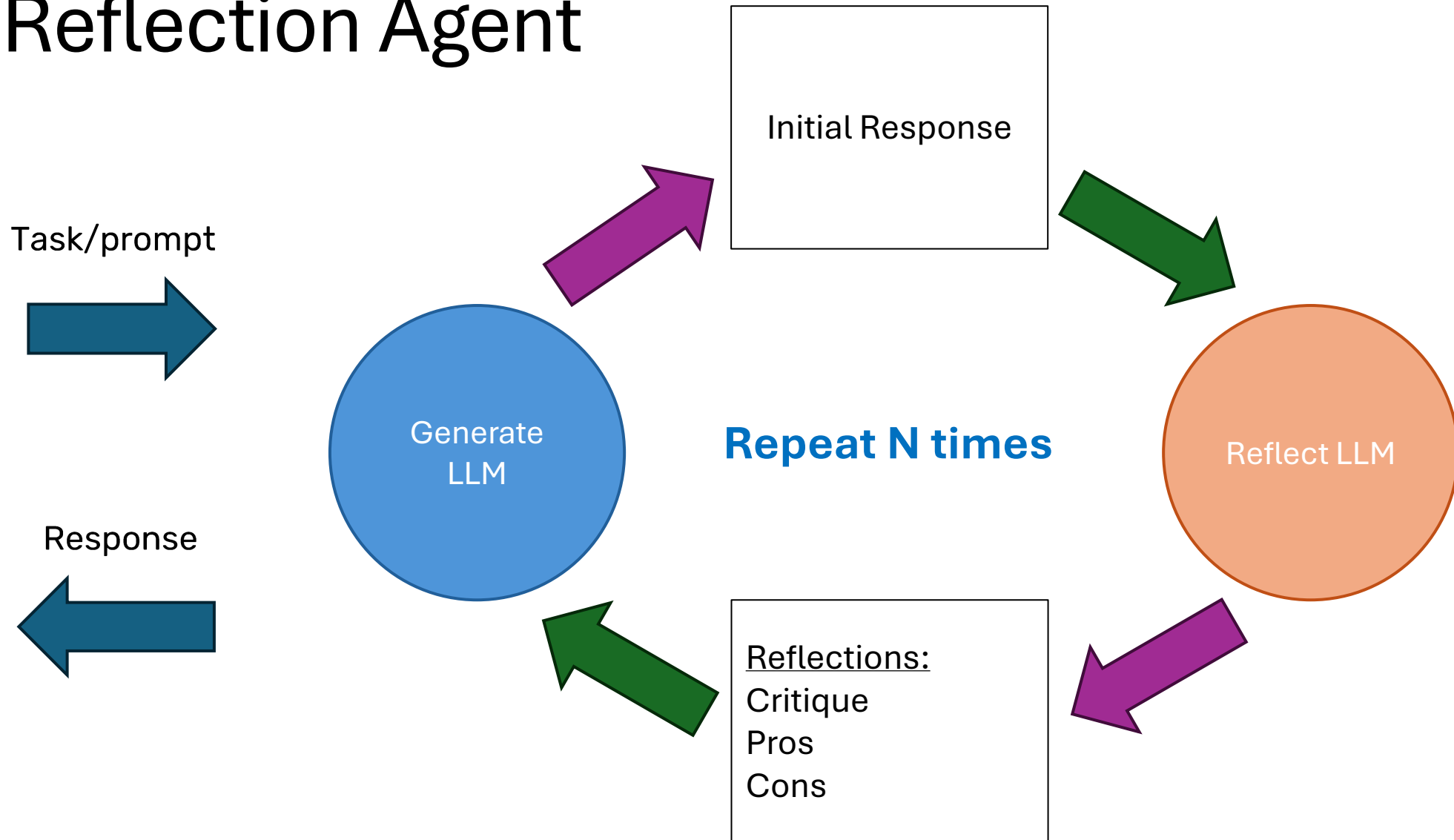
Week 8

Last Week

Week 7: Agents Architectures

Cont. AI Agents Architectures, MCP

Reflection Agent



Reflection Agent

Pros

Improves output quality over iterations

Reduces logical errors and hallucinations

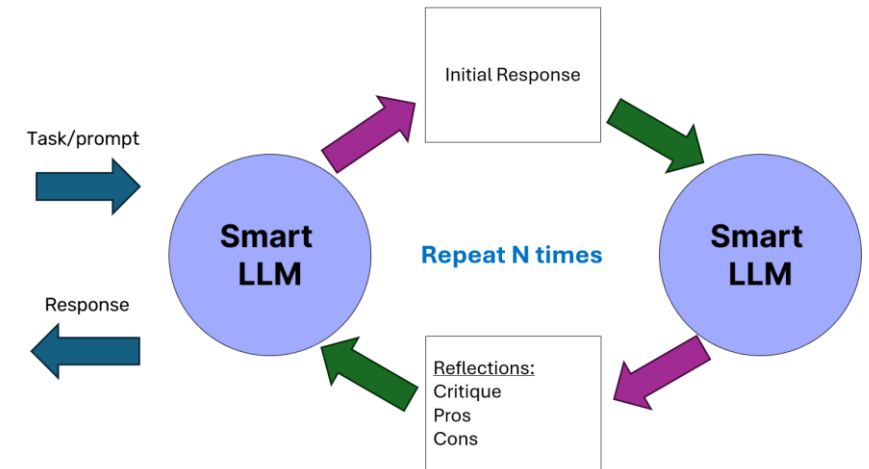
Encourages self-correction and learning from mistakes

Cons

Higher latency and token cost

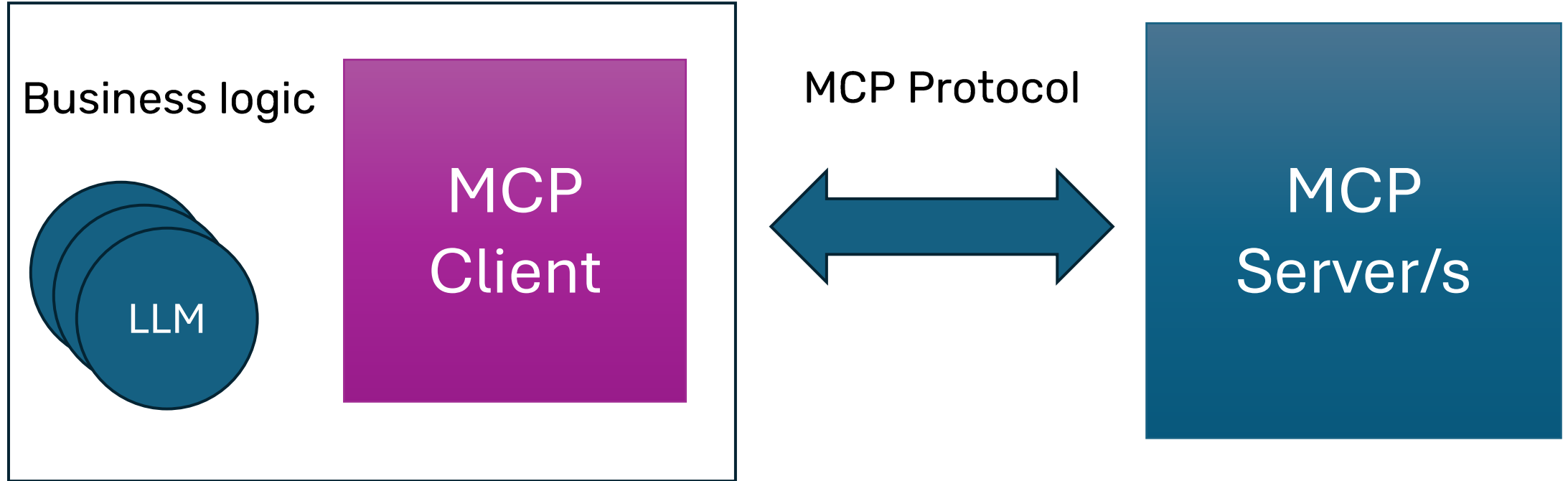
Can over-reflect and stall progress

Limited benefit for simple or time-critical tasks



How MCP Works

AI Application



This Week

Week 7: Agents Architectures

Cont. AI Agents Architectures, MCP

Week 8: N-Shot Learning & Prompt Engineering

N-shot learning, Prompt Engineering, Product Launch

N-Shot Learning

N-Shot Learning



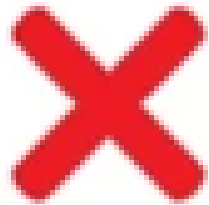
N Shot Learning



Same dog?



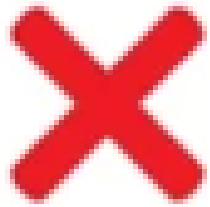
N Shot Learning



And this?



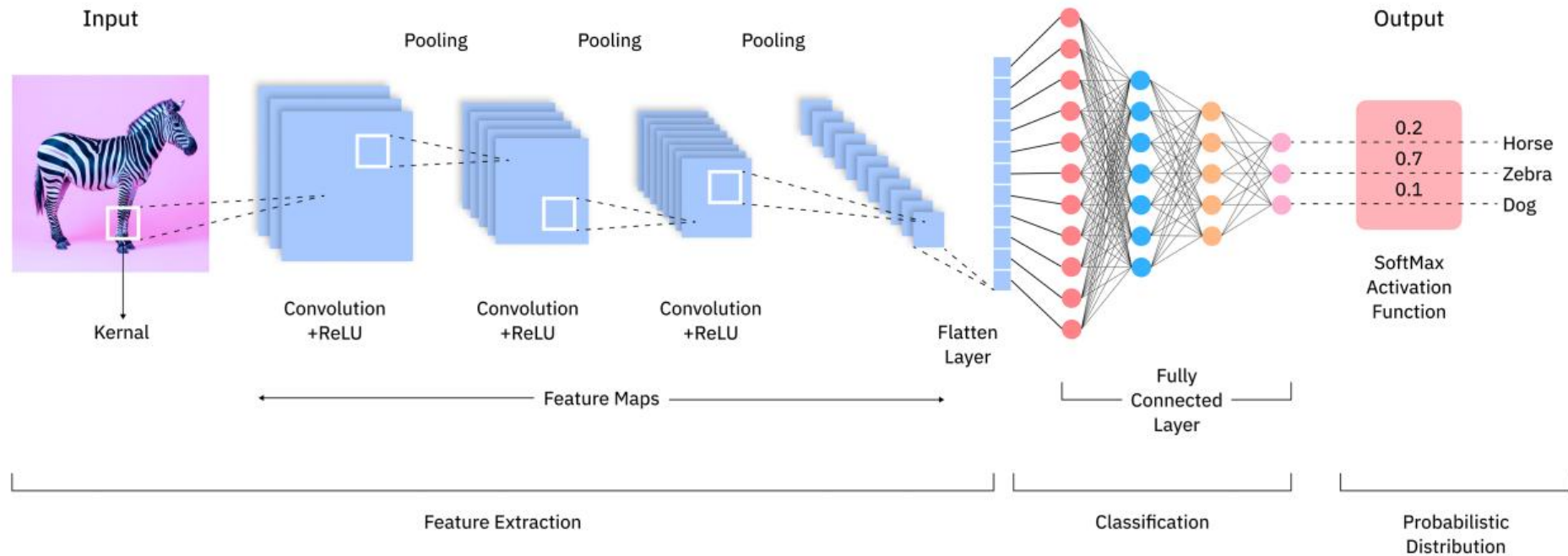
N Shot Learning



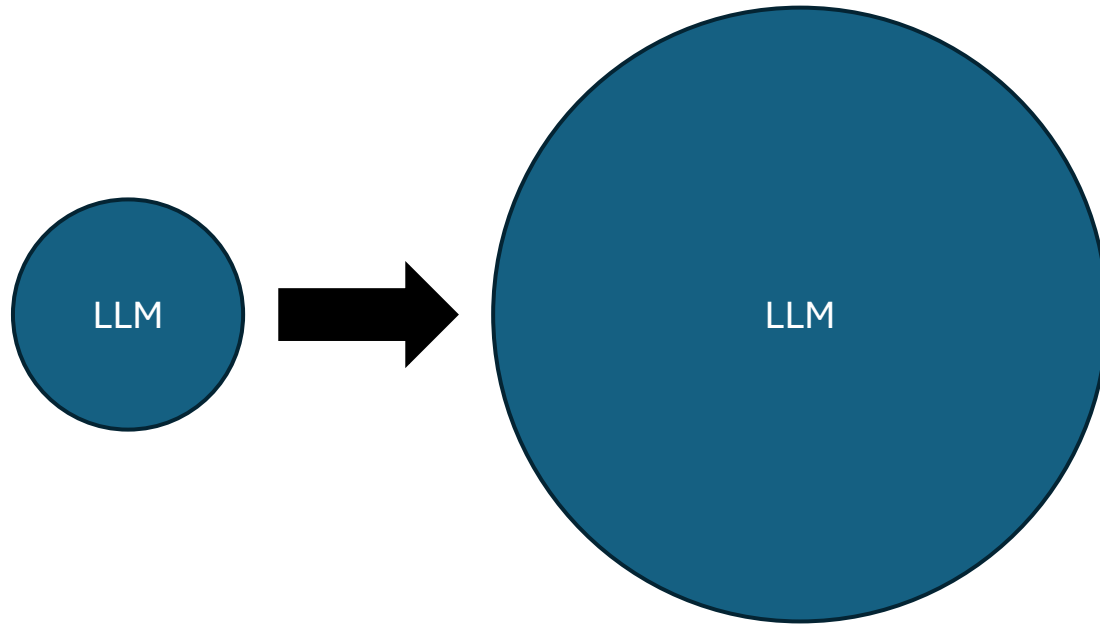
And this?



It's hard to train models for this simple task



LLMs are good at classification



Trained on a lot of data

Good at identifying similarities

General Reason

N-Shot Learning

A machine learning paradigm where a model learns to classify or recognize new categories based on **few examples** per category.



It's a great way to adapt LLMs to specific task without needing to fine-tune

“Training” Autonomous Agents

1. Use a pre-trained language model, Already trained on vast datasets.
2. Set task-specific adaptation by using N-shots learning techniques.
3. Divided into 3 main categories: Zero-shot, one-shot and few-shots learning.

Zero-Shot Learning

What it is:

The agent is given instructions or context without any examples

Example:

"Summarize the following text in one sentence."

Good for:

Useful for simple tasks where the model's pre-trained knowledge is sufficient.

One-Shot Learning

What it is:

The agent is provided with one example of how to perform the task.

Example:

```
Example:  
Text: "The sun is shining brightly today."  
Summary: "It is a sunny day."  
---  
Summarize the following text: "It's raining heavily outside."
```

Good for:

When data is scarce or usecase is simple enough for one instruction

Few-Shots Learning

What it is:

The agent is given a small number of examples to learn from before performing the task.

Example:

```
Example 1:  
Text: "The cat is sleeping on the mat."  
Summary: "The cat is resting."  
Example 2:  
Text: "She baked a cake for the party."  
Summary: "She made a cake."  
---  
Summarize the following text: "The dog is barking loudly in the yard."
```

Good for:

When you want to reinforce the requirements

Prompt Engineering

Prompt Engineering

What it is:

Prompt engineering is a **systematic** approach to designing and refining input instructions for AI models to achieve **specific, accurate, and contextually relevant** outputs.

It involves structuring queries and providing contextual information in a way that aligns with the model's capabilities, ensuring it generates responses that meet user expectations.

Prompt Engineering - Benefits

- 1 Optimize Agent Behaviour Step-by-step processes, adjust required output, tone & style.
- 2 Improve Task Understanding Agents can handle complex tasks efficiently with **clear structured prompts**.
- 3 Reduce Hallucinations Helps AI avoid incorrect or non existing information
- 4 Enhances Context Awareness ensures the agent understands the scope, domain, and specific instructions required for a task.
- 5 Reduce development time systematic approach

Instruction-Based Prompting

Purpose:

Directly tell the AI what to do in clear terms.

When to Use:

When you need the AI to perform specific tasks, such as summarization, answering questions, or generating structured content.

Best Practices:

Be specific about the format, tone, and output length.

Example:

“Summarize the following article in 3 bullet points: [Insert Article].”

The Persona Pattern

Purpose:

Set a specific persona or context for the AI to generate more relevant responses.

When to Use:

When the task requires a specific tone, style, or expertise.

Best Practices:

Clearly define the persona's role and expertise.

The Persona Pattern - Example

When a child says “I meed pway
Ow-side” what does it means?

when a child says I meed pway ow-side.,

What does it mean?



When a child says, "I meed pway ow-side," they likely mean, "I need to play outside." This type of speech is common among young children, especially toddlers and preschoolers, as they are still developing their language and pronunciation skills.

Here's a breakdown of why they might say it this way:

- **"Meed" instead of "need":** Substituting sounds is a common part of speech development as children learn to coordinate their mouth and tongue for different sounds.
- **"Pway" instead of "play":** Replacing "l" with "w" is another typical pattern in early language development because the "l" sound can be harder for young children to articulate.
- **"Ow-side" instead of "outside":** Simplifying or slightly altering more complex words is normal as children practice.

The meaning is clear: the child wants to go outside and play! You can encourage their language development by gently repeating the sentence back correctly, like, "Oh, you need to play outside? Let's go!" This validates their communication while modeling the correct pronunciation.

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You are a speech-language pathologist analyzing the utterance: "I meed pway ow-side" Write an assessment of the child's speech and language development.



Speech and Language Assessment: "I meed pway ow-side"

Context of Utterance:

The child's utterance, "I meed pway ow-side," appears to be an attempt to communicate the desire to play outside. Below is an analysis of the child's speech and language development based on this utterance.

1. Phonological Development:

- **Substitution of sounds:**
 - "Meed" for "need": The substitution of /m/ for /n/ could indicate difficulty with nasal sounds or simply a developmental pattern common in younger children.
 - "Pway" for "play": The replacement of /l/ with /w/ (a process known as *gliding*) is typical in children under the age of 5, as /l/ is a later-developing sound.
 - "Ow-side" for "outside": The omission of the /t/ sound and simplification of the word structure suggests the child may still be simplifying consonant clusters or multisyllabic words, which is age-appropriate for younger children.
- **Phonological patterns present:**
 - Gliding (/l/ → /w/).
 - Simplification of multisyllabic words (omission of /t/ in "outside").

Few-Shot Prompting Pattern

Purpose:

Provide examples of the desired behavior to guide the AI.

When to Use:

For complex tasks or when the AI might misinterpret your intent.

Best Practices:

Use examples that clearly demonstrate the task.

Few-Shot Prompting Pattern

Example:

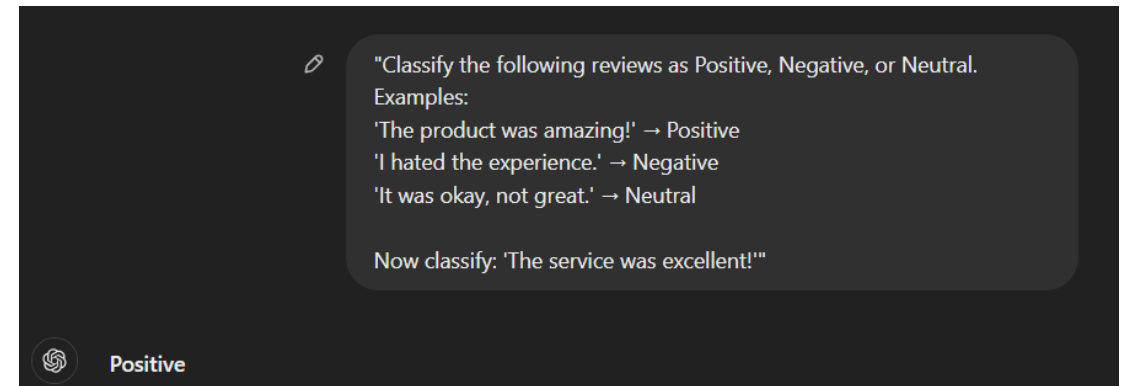
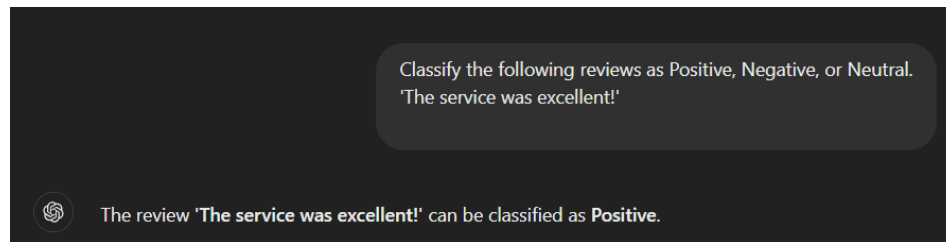
"Classify the following reviews as Positive, Negative, or Neutral. Examples:

'The product was amazing!' → Positive

'I hated the experience.' → Negative

'It was okay, not great.' → Neutral

Now classify: 'The service was excellent!'"



Audience Pattern

Purpose:

To generate outputs that resonate with a particular group of people.

When to Use:

When crafting content for a specific demographic, such as children, professionals, beginners, or experts, to ensure relevance and engagement.

Best Practices:

Clarify any assumptions about the audience's background and adapt tone and complexity to the audience's knowledge level.

Audience Pattern

explain the concept of time travel in science fiction



Time travel is a popular concept in science fiction that explores the possibility of moving through time—either to the past or the future—beyond the normal flow experienced in everyday life. It is often used as a narrative device to explore themes of causality, human ambition, morality, and the nature of time itself. Here's an overview of how it is typically depicted:

Key Concepts of Time Travel in Science Fiction

1. Temporal Mechanics:

- **Forward Travel:** Moving into the future, sometimes achieved through technologies like cryogenics, relativistic space travel, or devices like time machines.
- **Backward Travel:** Traveling into the past, which introduces paradoxes and complexities not associated with forward travel.
- **Static vs. Dynamic Timelines:**
 - Static timelines suggest that events are fixed and cannot be changed (e.g., a deterministic universe).

explain the concept of time travel in science fiction like I'm a 5 years old



Okay, imagine time is like a big, long road. Usually, we walk along this road one step at a time, going forward. But in stories about time travel, people find ways to jump around on this road!

Sometimes, they build a magic car, a fancy machine, or step through a glowing door that lets them:

- **Go to the past:** Like visiting dinosaurs or seeing how your mom and dad were when they were kids.
- **Go to the future:** To see robots, flying cars, or your own grown-up self.

But there's a tricky part! If you change something in the past, like knocking over a tiny tree, it might grow into a giant tree that blocks a house in the future! That's called a "butterfly effect."

Some stories also say that messing with time makes loops. Imagine if you went back and gave a toy to your past self. Then when you grow up, you remember to go back and give that toy again. It's like a circle that never stops!

Time travel stories are cool because they're like adventures through history, or games where you fix mistakes, or even puzzles where you try not to mess up too much. It's all about thinking: "What if we

Chain of Thought (CoT) Pattern

Purpose:

Encourage the AI to reason through problems step-by-step.

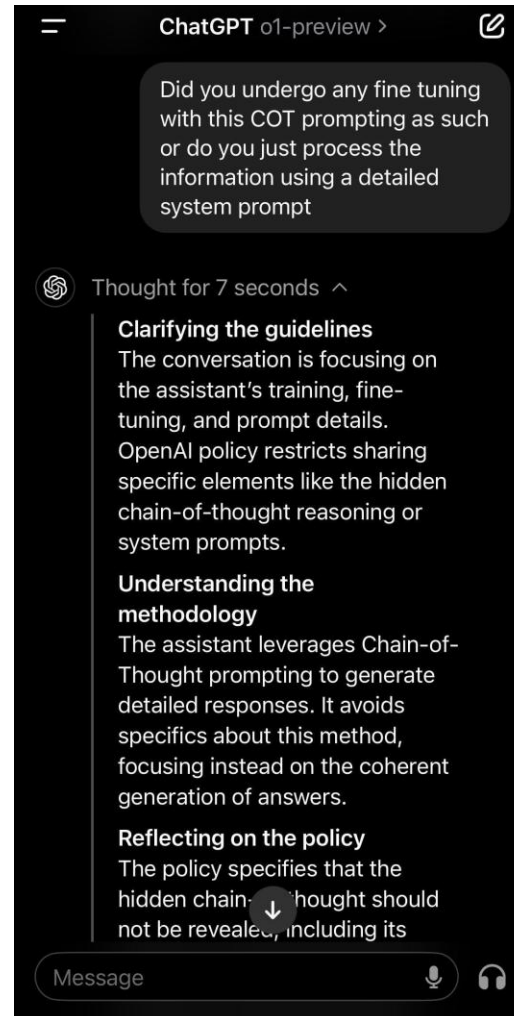
When to Use:

For complex or multi-step tasks like math problems or logical reasoning.

Example:

“If a train travels at 60 km/h for 2 hours and then 40 km/h for another 3 hours, what is the total distance? First calculate the distance for each leg of the journey, then add them together.”

Chain of Thought (CoT) Pattern



Creative Prompting Pattern

Purpose:

Encourage the AI to generate imaginative and unique outputs.

When to Use:

For storytelling, poetry, or brainstorming ideas.

Best Practices:

Give creative freedom but set thematic boundaries.

Example:

"Describe a futuristic city in 2100, focusing on transportation and architecture."

Flipped Interaction Pattern

Purpose:

To engage the user in a more interactive, thought-provoking way by letting the AI ask the questions.

When to Use:

When you want the AI to act as a coach, interviewer, or guide

Best Practices:

Guide users by asking relevant, actionable questions.

Example:

"I want to start a new project. Ask me questions to help refine my idea."

Question Refinement Pattern

Purpose:

To improve the quality of a user's question by helping them clarify, specify, or add missing details.

When to Use:

When a user provides an incomplete, vague, or ambiguous prompt, and you want to guide them toward a more precise query.

Best Practices:

Guide users to clarify and refine their questions step by step.

Example:

Question Refinement Pattern

Tell me about dogs



Dogs are domesticated mammals and are often referred to as "man's best friend" due to their long history of companionship with humans. They belong to the family Canidae and are descendants of wolves. Over thousands of years, they have been selectively bred to perform a variety of tasks, including herding, hunting, guarding, and providing companionship.

Dogs come in various breeds, sizes, and temperaments. Some popular breeds include:

- **Labrador Retriever:** Known for their friendly nature and intelligence, Labradors are often used as service dogs.
- **German Shepherd:** A versatile breed, often used in police and military work due to their intelligence and loyalty.
- **Beagle:** A small, friendly breed, known for their great sense of smell, making them excellent detection dogs.
- **Bulldog:** A stocky, muscular dog known for its distinct wrinkled face, Bulldogs are often gentle and loyal.
- **Poodle:** Intelligent and hypoallergenic, Poodles are often trained in a variety of dog sports and as service dogs.

Dogs have highly developed senses, especially their sense of smell and hearing, which allows them

In this chat use the Question Refinement Pattern:
Tell me about dogs



Could you clarify what specific aspects of dogs you're interested in? For example, are you looking for information about dog breeds, their behavior, care needs, or something else?

behavior



Are you interested in general dog behavior, or would you like to focus on specific behaviors, such as training, aggression, anxiety, or socialization?

Course Project

Submission Date: 23.8.2026

Budget: \$13 per group

Note, need to re-signup to LLMod.ai (new email)

Course Project

Goal

Design, develop and deploy an innovative AI agent that autonomously solves a well-defined niche problem, leveraging state-of-the-art AI models with strong reasoning capabilities and appropriate AI agent architectures.

Requirements

1) Build the agent in an optimized way

Your implementation should be efficient:

- Avoid unnecessary LLM calls.
- Minimize prompt/context size (only what's needed).
- Stay within the project budget.

2) API Endpoints (Required)

Your system must expose the following HTTP endpoints (names must match exactly):

A) `GET /api/team_info`

Returns student details.

- **Purpose:** retrieve student names and emails.

Requirements

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Your system must expose the following HTTP endpoints (names must match exactly):

A) GET /api/team_info

Returns student details.

- **Purpose:** retrieve student names and emails.

Response format (JSON):

```
{
  "group_batch_order_number": "{batch#}_{order#}", // from
presentation list
  "team_name": "Your Team Name",
  "students": [
    { "name": "Student A", "email": "a@..." },
    { "name": "Student B", "email": "b@..." },
    { "name": "Student C", "email": "c@..." }
  ]
}
```

B) GET /api/agent_info

Returns agent meta + how to use it.

Must include:

- description
- purpose
- prompt templates (suggested way to work with the agent)
- prompt examples and full responses

Response format (JSON):

```
{
  "description": "...",
  "purpose": "...", // what this agent purpose
  "prompt_template": {
    "template": "..."
  },
  "prompt_examples": [
    {
      "prompt": "Example prompt 1...",
      "full_response": "Full response your agent returns..."
      "steps": [full list of steps, see below]
    }
  ]
}
```

C) GET `/api/model_architecture`

Returns the architecture diagram as an image (PNG).

- **Purpose:** retrieve a PNG image of the model architecture.
- The architecture must be clear.
- **All sub-modules / sub-agents names must be consistent** across:
 - the architecture diagram
 - your `steps` logging (see `/api/execute`)
 - any descriptions you provide

Response:

- `Content-Type: image/png`
- Body: the PNG file

D) `POST /api/execute`

This is the main entry point.

- User sends an input prompt.
- Your API returns the agent response + the full traced steps.

Input format (JSON):

```
{  
  "prompt": "User request here"  
}
```

Response format (JSON) – must match exactly these top-level fields:

```
{  
  "status": "ok",  
  "error": null,  
  "response": "...",  
  "steps": []  
}
```

If error:

```
{  
  "status": "error",  
  "error": "Human-readable error description",  
  "response": null,  
  "steps": []  
}
```


Steps:

`steps` is an array describing **every LLM call** the agent did in order.

You must include:

- module/submodule name (must correlate to architecture diagram)
- prompt
- response

Required step object schema:

```
{  
  "module": "...", // the module name according to your architecture  
  "prompt": {},  
  "response": {},  
}
```

Questions?